Project: Worked examples
Title: Rolling bearing basic rating life (simple form)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the equivalent dynamic load, the basic rating life L_{10} and the life modified for reliability of a rolling bearing, against a required life. It takes the reliability factor as entered and has no a_ISO factor.

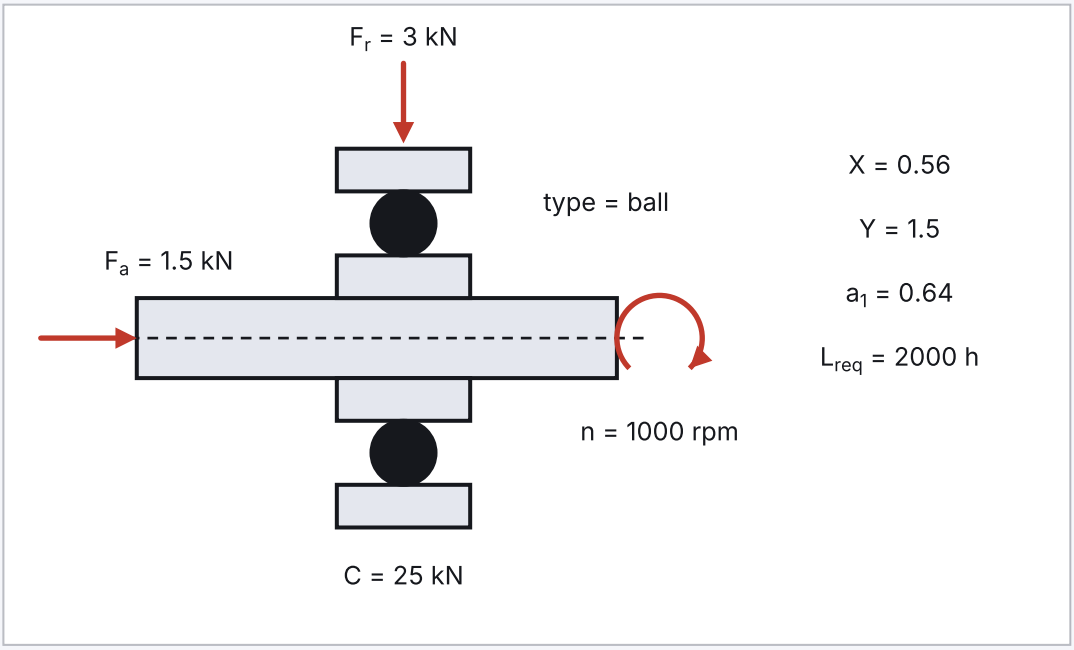


Figure 1 — A ball bearing on its shaft in section, radial and axial load, running speed n. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Equivalent dynamic load from the radial and axial loads	$F_{dyn} = X \cdot F_r + Y \cdot F_a$
Basic rating life in millions of revolutions (p = 3 for ball and 10/3 for roller bearings)	$L_{10} = (C/F_{dyn})^p$
Rating life in hours, at speed n	$L_{10h} = L_{10} \cdot 10^6$ revolutions at speed n
Life adjusted for the required reliability	$L_{nm} = a_1 \cdot L_{10}$; $L_{nmh} = a_1 \cdot L_{10h}$
Range of validity of the life equation	$F_{dyn} \le 0.5 \cdot C$

Basis: Basic dynamic load rating and basic rating life[1](cl. 6); Rolling-contact bearings[2](Ch. 11, 11th ed.).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
type	Bearing type (selects the life exponent)	ball		drawing: DRG-GB-211 rev A	any
C	Basic dynamic load rating	25	kN	drawing: DRG-GB-211 rev A	0.1 … 100000 kN
F_r	Radial load	3	kN	spec: LOAD-SCHEDULE-01	0 … 100000 kN

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
F_a	Axial load	1.5	kN	spec: LOAD-SCHEDULE-01	0 … 100000 kN
n	Speed	1000	rpm	spec: LOAD-SCHEDULE-01	1 … 100000 rpm
X	Radial load factor (bearing catalogue)	0.56		drawing: DRG-GB-211 rev A	0 … 2
Y	Axial load factor (bearing catalogue)	1.5		drawing: DRG-GB-211 rev A	0 … 5
a_1	Life modification factor for reliability	0.64		spec: SPEC-MECH-03 design criteria	0.05 … 1
L_{req}	Required life (0 for no criterion)	2000	h	spec: SPEC-MECH-03 design criteria	0 … 1000000 h

3. Assumptions

A1. Load and speed are steady, and the bearing is correctly fitted, lubricated and free of contamination. (These are the conditions of the basic rating life.)
A2. X and Y are the catalogue factors for this bearing and load ratio. (Supplied by the user.)

4. Working

[10] p = 3 [1](281:2007 cl. 6)
= 3
Life exponent (ball bearing)

[11] F_{dyn} = X * F_r + Y * F_a [1](281:2007 cl. 6)
= 0.56 × 3 kN + 1.5 × 1.5 kN
= 3.93 kN
Equivalent dynamic load

[12] k_P = F_{dyn} / C
= 3.93 kN / 25 kN
= 0.1572
Load ratio of the equivalent load to the rating

[13] L_{10} = (C / F_{dyn})^p [1](281:2007 cl. 6)
= (25 kN / 3.93 kN)^3
= 257.42
Basic rating life (millions of revolutions)

[14] L_{10h} = L_{10} * 1e6 * 2 * pi / n
= 257.42 × 1e6 × 2 × π / 1000 rpm
= 4290.3 h
Basic rating life in hours (10^6 revolutions of 2π rad at speed n)

[15] L_{nm} = a_1 * L_{10}
= 0.64 × 257.42
= 164.75
Modified rating life (millions of revolutions)

[16] L_{nmh} = a_1 * L_{10h}
= 0.64 × 4290.3 h
= 2745.8 h
Modified rating life in hours

[17] $u_{life} = L_{req} / L_{nmh}$
= 2000 h / 2745.8 h
= 0.72838
Required over modified life

5. Results

SYMBOL	VALUE	DESCRIPTION
L_{nmh}	2746 h	Modified life in hours
L_{10h}	4290 h	L_{10} in hours
F_{dyn}	3.930 kN	Equivalent dynamic load
k_P	0.1572	Load ratio
L_{10}	257.42	L_{10} (millions of revolutions)
L_{nm}	164.75	Modified life (millions of revolutions)
u_{life}	0.73	Required over modified life (governing)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Required life against modified life	2000 h	2746 h	0.73	[1](281:2007 cl. 6)

7. Applicability

None.

8. Verification

Template mech.bearing-life-basic@1.1.0 Content hash 3a6e1cc22fcb1adc Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. International Organization for Standardization. Rolling Bearings—Dynamic Load Ratings and Rating Life. ISO 281. ISO.
2. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.